



SOMAIYA
VIDYAVIHAR

K J Somaiya Institute of Technology
An Autonomous Institute permanently affiliated to University of Mumbai.

K J SOMAIYA INSTITUTE OF TECHNOLOGY, SION, MUMBAI

DEPARTMENT OF COMPUTER ENGINEERING

Snap & Report - Civic Issue Reporting

SY Mini Project Report

**Submitted in partial fulfillment of the requirements
of the Subject Project Based Learning: Mini Project Lab-I (Sem III)**

by

**Atharva Naik (B-2)
Vikas Patel (B-14)
Arin Sutariya (B-48)
Anshuman Tiwari
(B-51)**

Supervisor

Mr. Guide Name



**Department of Computer
Engineering K J Somaiya Institute of
Technology Ayurvihar, Sion
Mumbai-400022 2025-26**



CERTIFICATE



*This is to certify that the project entitled “**Project Title**” is bonafide work **Student Name** submitted as a SY Sem III Mini project in the subject of Project Based Learning: Mini Project Lab-II, Computer Engineering for the academic year 2025-26.*

Guide Name
Project Guide
Department of Computer Engineering

Dr. Sarita Ambadekar
Head of Department
Dept. of Computer Engineering

Dr. Vivek
Sunnapwar
Principal
KJSIT, Sion

Place: Sion, Mumbai 400022

Date:

PROJECT APPROVAL FOR S. Y.

This project report entitled “**Hand Gesture Controlled Presentation**”

Rhushabh J. Pradhan – B/14

Raj K. Sanghvi– B/25

Aaditya A. Verma – B/54

is an approved Second Year Mini Project **in Computer Engineering.**

EXAMINER:

1. _____

External Examiner Name and Sign

2. _____

Internal Examiner Name and Sign

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Student Name _____

(sign of student)

Student Name _____

(sign of student)

Student Name _____

(sign of student)

Date:

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Place: Sion, Mumbai-400022

Date:

ABSTRACT

In today's world, as businesses digitize and rely on cloud computing, the risk of cyberattacks rises. Malware infections, denial-of-service, and phishing are among the many types of attacks. Therefore, safeguarding data integrity and confidentiality and preventing harm to computer systems and networks through security measures is critical. An Intrusion Detection System (IDS) monitors network traffic to detect unauthorized access attempts. It can use two methods: signature-based, which identifies known threats based on pre-determined patterns, and anomaly-based, which analyzes traffic for unusual behavior. Anomaly-based detection can detect emerging threats, but it may also generate false positives and false negatives. To overcome the limitations of each method, a hybrid IDS approach has been proposed. This approach combines the benefits of both signature-based and anomaly-based detection while minimizing their downsides. In a hybrid IDS, known threats are detected using signature-based detection, while previously unknown or emerging threats are detected using anomaly-based detection. A study introduced a hybrid IDS utilizing autoencoder-based anomaly detection and signature-based detection based on Random Forest and Light GBM classification models. The CICID2017 dataset, a standard IDS research benchmark dataset, was preprocessed, and autoencoders reduced its dimensions. The authors then trained the classification models using Random Forest and Light GBM, two commonly used algorithms for classification tasks.

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LIST OF ABBREVIATIONS

Sr. No	Abbreviation	Description
1	IDS	Intrusion Detection System
2		
3		
4		
5		
6		
7		

CHAPTER 1

INTRODUCTION

In today's fast-paced world, civic maintenance plays a vital role in ensuring public safety, comfort, and quality of life. However, many urban and suburban areas continue to face persistent neighborhood problems such as potholes, broken streetlights, overflowing garbage bins, water leakage, and damaged public infrastructure. These issues, though common, are often neglected due to inefficient communication between citizens and local authorities.

Traditionally, the process of reporting civic problems involves visiting municipal offices, making phone calls, or submitting written complaints. Such methods are often slow, inconvenient, and lack transparency. As a result, citizens become discouraged from reporting issues, leading to prolonged neglect and further deterioration of public spaces. Moreover, authorities frequently remain unaware of these problems until they escalate, due to the absence of a proper and timely reporting system.

To address these challenges, **Snap & Report** is proposed as a modern, technology-based solution that simplifies and accelerates the process of reporting civic issues. The system enables users to capture images of problems such as potholes, damaged roads, or garbage piles, along with their location details, and directly submit them to the concerned authorities. This approach not only enhances communication and accountability but also promotes greater citizen participation in maintaining the cleanliness and safety of their surroundings.

By integrating features such as real-time complaint submission, location tracking, and status updates, **Snap & Report** aims to create a transparent and efficient system for issue management. The platform encourages collaboration between citizens and civic bodies, ensuring that local problems are identified, tracked, and resolved promptly. Ultimately, the project strives to contribute to smarter, cleaner, and safer communities through the effective use of technology.

1.1 Problem Definition

The proposed system, **Snap & Report**, is designed to provide an efficient, technology-driven platform for addressing everyday neighborhood issues faced by citizens. In many communities, civic problems such as potholes, broken streetlights, damaged roads, garbage accumulation, and water leakages are common and often go unnoticed or unresolved for long periods. These issues not only affect the convenience and comfort of residents but also pose risks to public safety, environmental hygiene, and the overall quality of life.

At present, the existing methods for reporting such problems—such as visiting municipal offices, making phone calls, or submitting written complaints—are time-consuming and inefficient. These outdated systems often discourage citizens from taking initiative because they involve lengthy procedures, lack acknowledgment, and offer little assurance of resolution. Moreover, many residents are unaware of how to properly report an issue, while others are unable to do so due to the absence of a quick and accessible system.

The lack of a simple, user-friendly mechanism for citizens to directly communicate with civic authorities is a major shortcoming in the current setup. There is no digital platform that allows users to instantly capture an image of a problem, specify its location, and submit it to the concerned department in real time. As a result, authorities often remain unaware of such issues until they escalate into more serious concerns.

Another critical challenge is the **absence of transparency and tracking** in existing complaint-handling processes. Citizens rarely receive updates about the status of their complaints, leading to frustration, lack of accountability, and diminished trust in public systems. This communication gap between residents and authorities results in delays in problem resolution and inefficient allocation of resources.

Unresolved civic issues can have significant negative consequences. Potholes can cause vehicle damage and accidents, broken streetlights can increase crime risk, and uncollected garbage can lead to health hazards. Collectively, these problems degrade the living conditions of residents and weaken public confidence in local governance. Hence, there is an urgent need for a smart, transparent, and user-friendly solution like **Snap & Report** to streamline the reporting process, ensure timely action, and improve overall civic management efficiency.

1.2 Aim and Objective

Aim

The main aim of the project “**Snap & Report**” is to develop a user-friendly and efficient digital platform that enables citizens to report civic issues such as potholes, broken streetlights, garbage accumulation, and other neighborhood problems directly to the concerned authorities in real time. The system aims to improve communication, enhance transparency, and promote timely resolution of local issues through the use of technology.

Objectives

The primary objectives of this project are as follows:

1. **To design and develop** an easy-to-use mobile and web-based application that allows users to capture images and report civic issues with precise location details.
2. **To enable real-time communication** between citizens and municipal authorities for faster identification and resolution of problems.
3. **To implement a transparent tracking system** that allows users to monitor the status and progress of their complaints.
4. **To encourage citizen participation** by simplifying the process of reporting and reducing the effort required to file complaints.
5. **To support local authorities** in efficiently managing and prioritizing civic issues based on frequency, severity, or location.
6. **To enhance accountability** within public departments by ensuring that every complaint is documented and tracked until resolution.
7. **To contribute towards building smarter and cleaner cities** through the effective use of technology and community engagement.

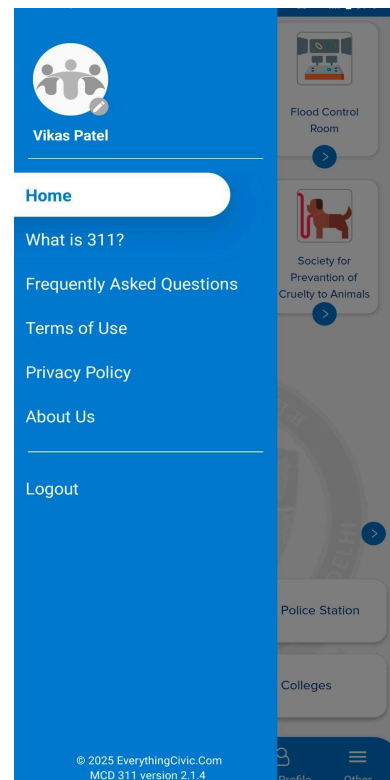
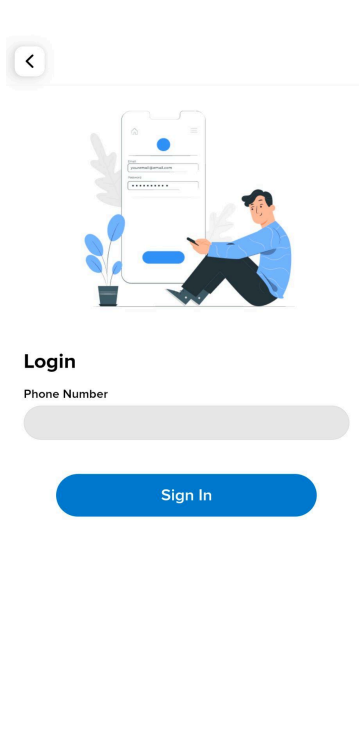
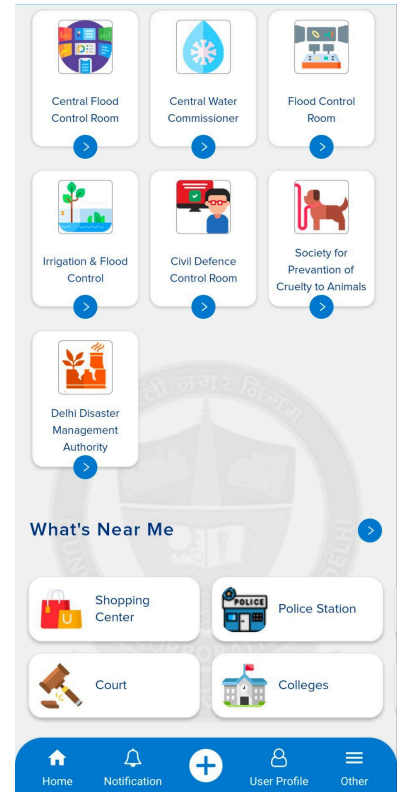
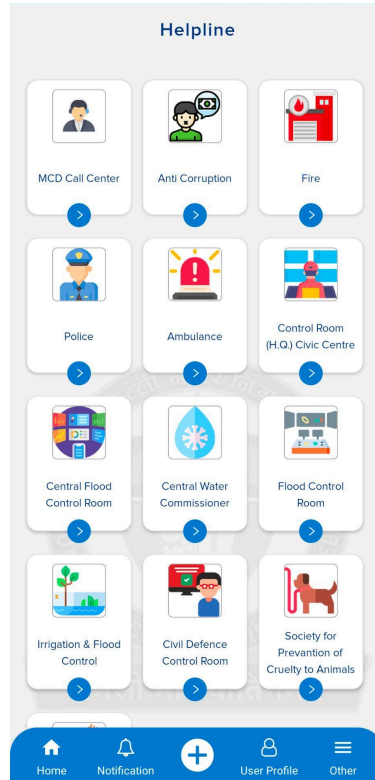
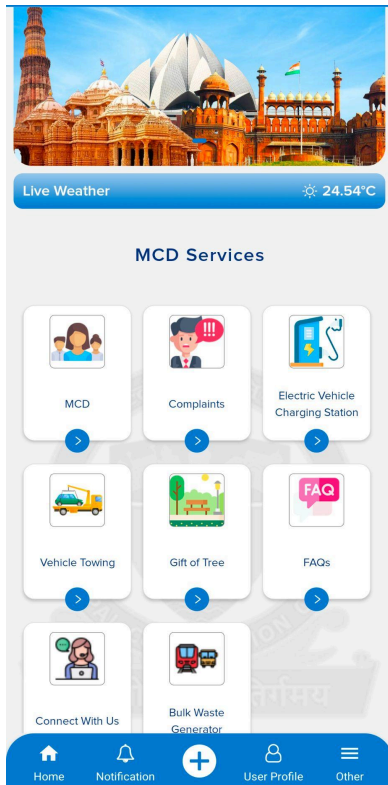
Organization of Report

This project report is organized into several chapters, each describing different aspects of the proposed system — “**Snap & Report.**” The structure of the report is as follows:

- **Chapter 1 – Introduction:**
This chapter provides an overview of the project, highlighting the background, aim, objectives, and the need for a digital solution to report civic issues efficiently.
- **Chapter 2 – Literature Review / Existing System:**
This chapter discusses the existing methods used for reporting neighborhood problems, their limitations, and the need for an improved, technology-based solution. It also includes a comparison between the existing and proposed systems.
- **Chapter 3 – System Analysis:**
This chapter explains the system requirements, feasibility study, and problem analysis. It identifies functional and non-functional requirements that guide the system design.
- **Chapter 4 – System Design:**
This chapter presents the detailed design of the proposed system, including architecture diagrams, data flow diagrams (DFDs), entity-relationship (ER) diagrams, and module descriptions.
- **Chapter 5 – System Implementation:**
This chapter describes the technologies, tools, and frameworks used in developing the “Snap & Report” system. It also explains how the modules are integrated and tested.
- **Chapter 6 – Testing and Results:**
This chapter includes the testing methods applied, test cases, and the results obtained. It also evaluates the system’s performance and efficiency.
- **Chapter 7 – Conclusion and Future Scope:**
This chapter concludes the project by summarizing the key outcomes and discusses possible enhancements or future developments for improving the system further.

CHAPTER 2

LITERATURE SURVEY



User Reviews

Overall Rating:

The app has received a very low average rating of 1.6 out of 5 stars, indicating widespread dissatisfaction among users.

Most users have given 1-star ratings, while very few rated it above 3 stars.

Common Complaints:

Frequent technical issues and bugs:

Users reported that when they open the complaint section or try to view existing complaints, a blank screen appears and nothing is displayed. This makes the app nearly unusable.

(Example: “Whenever I go to the complaint section and click on view all complaints, a blank screen comes up and nothing is visible.” — Pushkar Bhati)

Slow performance and poor response time:

Several users mentioned that the app is very slow even with a good internet connection, and that the work process is delayed.

(Example: “The app is very slow, even with high-speed internet, and the work process is just as delayed.” — KnorFix)

Inaccurate complaint updates:

Some users said that even after their issue was resolved, the app continued to show the complaint as “unresolved,” or it was marked “resolved” using unrelated or fake photos.

(Example: “They marked it as resolved with a random photo from another place.” — KnorFix)

Jurisdiction and registration errors:

Users faced difficulties while registering complaints, especially when the system displayed jurisdiction-related errors even for valid areas.

(Example: “When I try to register a complaint on an app, it says ‘you are not in jurisdiction of Jaipur heritage’ even though my area is under it.” — Anuj Gupta)

User Sentiment:

Users expressed frustration and disappointment with the app’s functionality, slow response, and lack of reliability.

Many felt that the system failed to serve its purpose of providing an easy way to report civic issues.

Progress Report

